Thampi et al. 2017

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Model Setup

```
closed.coral.model1 <- function (t, y, params) {
  ## first extract the state variables
  M <- y[1]
  C <- y[2]
  P <- y[3]
  X <- y[4]
  #TT <- y[5]

  ## now extract the parameters
  alpha <- params["alpha"]
  gamma <- params["gamma"]
  r <- params["r"]
  d <- params["d"]
  kappa <- params["kappa"]
  sigma <- params["sigma"]
  s <- params["s"]
  j <- params["j"]
  phi <- params["phi"]
  TT <- 1 - M - C

  ## now code the model equations
  dMdt <- alpha*M*C - (P*M)/(M+TT) + gamma*M*TT
  dCdt <- r*TT*C - d*C - alpha*M*C
  #dTTdt <- (P*M)/(M+TT) - gamma*M*TT - r*TT*C + d*C
  dPdt <- s*P*(1-(P/C)) - sigma*P*(1-X)
  dXdt <- kappa*X*(1-X)*(-1+j*(1-C)-sigma*P*(1-X)+phi *(2*X-1))

  ## combine results into a single vector
  dydt <- c(dMdt,dCdt, dPdt, dXdt)
  ## return result as a list!

  list(dydt)
}

closed.coral.model2 <- function (t, y, params) {
  ## first extract the state variables
  M <- y[1]
  C <- y[2]
```

```
  P <- y[3]
  X <- y[4]
  #TT <- y[5]

  ## now extract the parameters
  alpha <- params["alpha"]
  gamma <- params["gamma"]
  r <- params["r"]
  d <- params["d"]
  kappa <- params["kappa"]
  sigma <- params["sigma"]
  s <- params["s"]
  j <- params["j"]
  phi <- params["phi"]
  z <- params["z"]
  TT <- 1 - M - C

  ## now code the model equations
  dMdt <- alpha*M*C - (P*M)/(M+TT) + gamma*M*TT
  dCdt <- r*TT*C - d*C - alpha*M*C
  #dTTdt <- (P*M)/(M+TT) - gamma*M*TT - r*TT*C + d*C
  dPdt <- s*P*(1-(P/(1-z*C))) - sigma*P*(1-X)
  dXdt <- kappa*X*(1-X)*(-1+j*(1-C)-sigma*P*(1-X)+phi *(2*X-1))

  ## combine results into a single vector
  dydt <- c(dMdt,dCdt, dPdt, dXdt)

  ## return result as a list!
  list(dydt)
}
```

Graphs included in paper

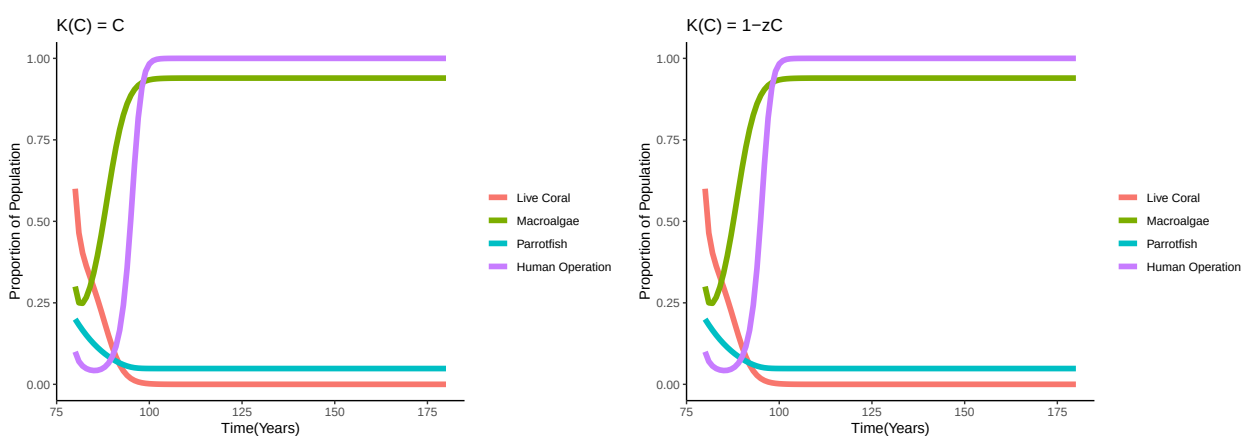

Figure 1: Fig 3a in Thampi paper. S = 0, sigma = 0.1

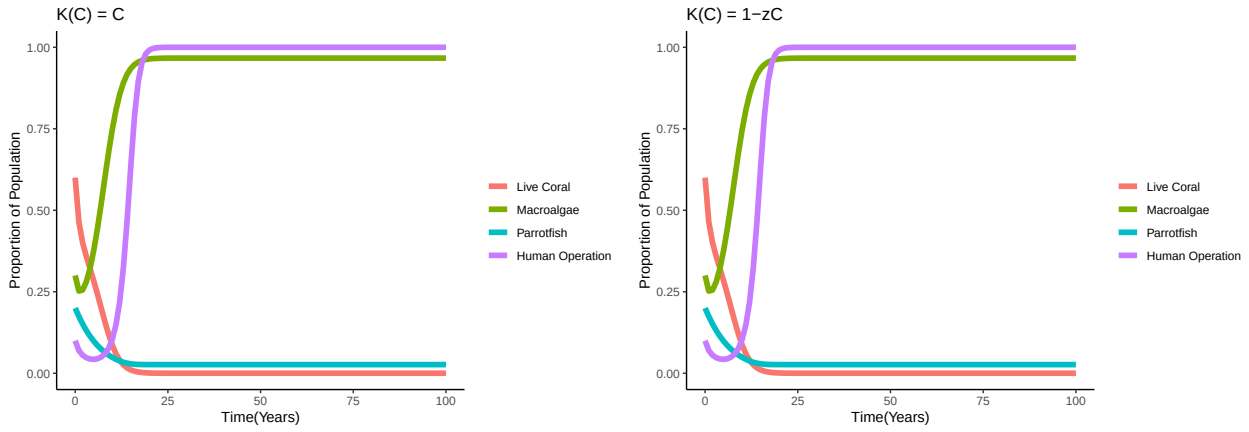

Figure 2: Fig 3b in Thampi paper. S = 0, sigma = 0.15

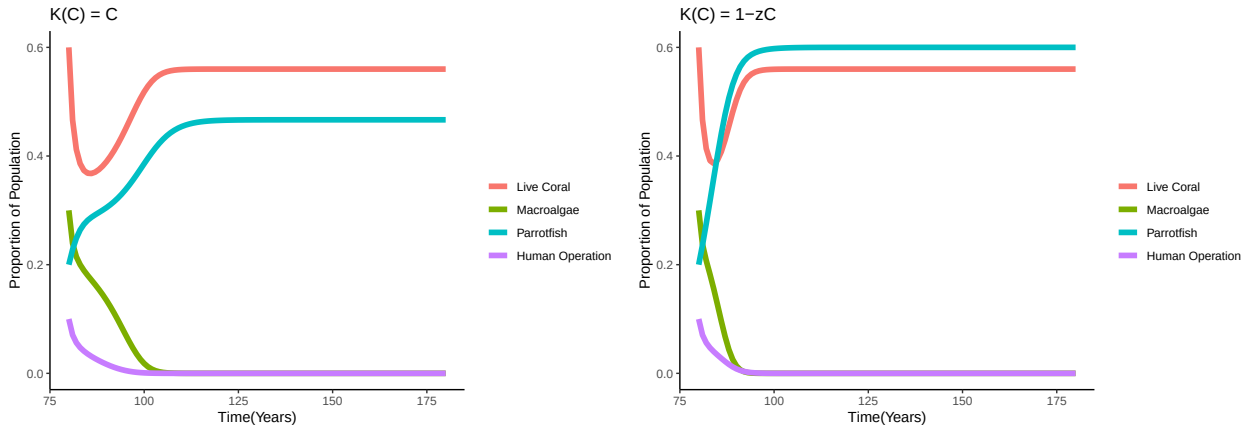

Figure 3: Fig 3c in Thampi paper. S = 0.3, sigma = 0.05

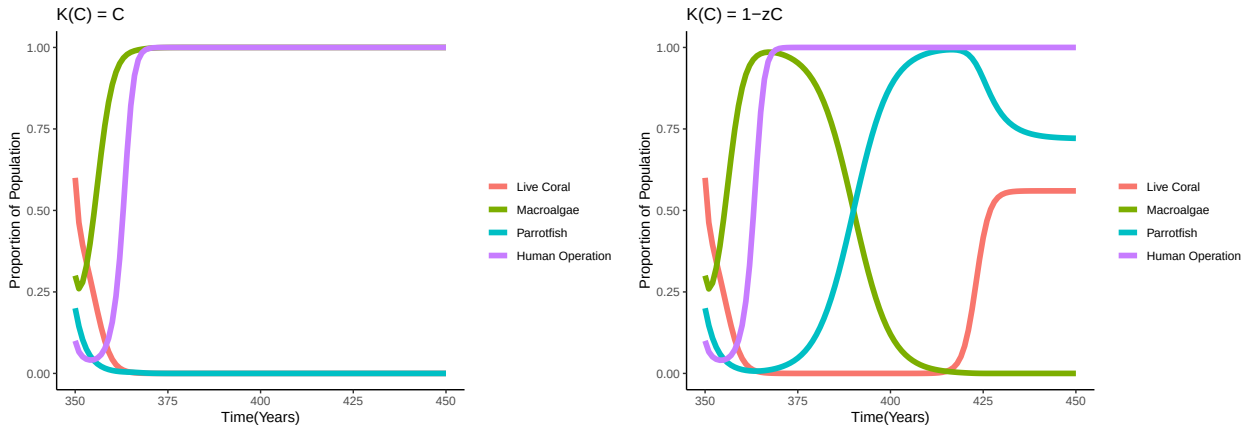

Figure 4: Fig 3d in Thampi paper. S = 0.2, sigma = 0.5

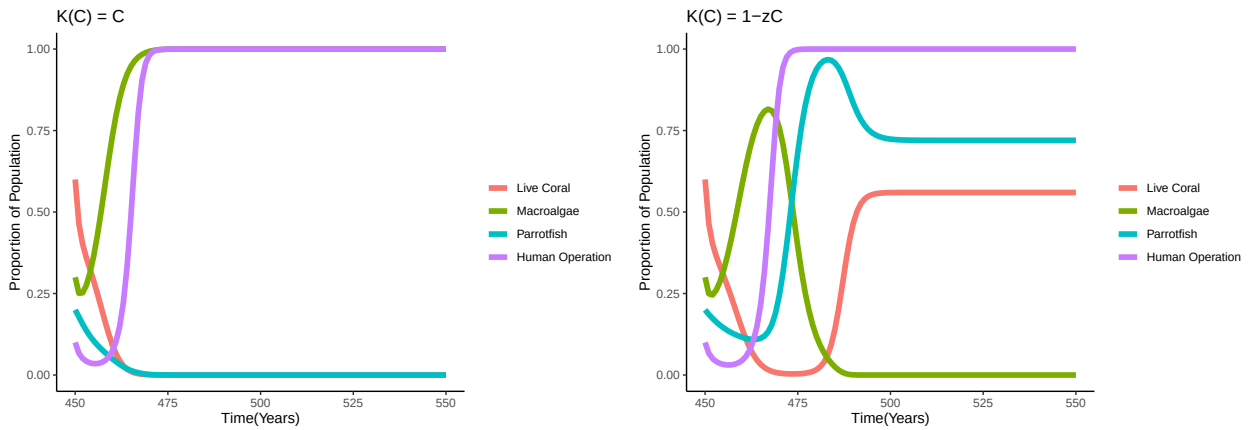

Figure 5: Fig 3e in Thampi paper. S = 0.4, sigma = 0.4

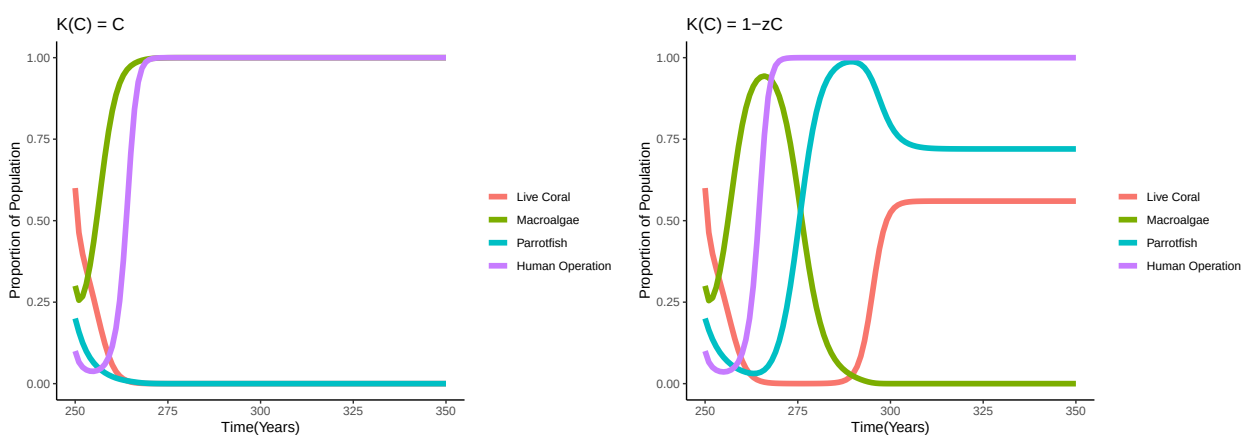

Figure 6: Fig 3f in Thampi paper. S = 0.35, sigma = 0.5